
The Attribution Impossibility:

No Importance Ranking Is Faithful, Stable, and Complete Under Symmetry

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Attribution instability is a mathematical property of statistical dependence—and
2 standard diagnostics are blind to it. We prove that no attribution ranking can simul-
3 taneously be faithful, stable, and complete when $I(X_j, X_k) > 0$ (the Attribution
4 Impossibility), with a quantitative floor: any stable explanation has error $\geq \Delta/2$
5 on at least one model. For binary questions, faithful + stable alone is impossi-
6 ble (the biletta). Standard multicollinearity diagnostics (VIF, Pearson) miss the
7 nonlinear cases entirely: on drug discovery data, Pearson predicts zero instabil-
8 ity for a pair with 23% actual instability. The theorem applies to *any* attribution
9 system: input-level (which feature matters) and component-level (which circuit
10 computes). **Input-level:** 50 retrains produce 24 distinct top-3 rankings (4.2%
11 agreement); 45% of German Credit applicants receive different explanation cat-
12 egories; 68% of 77 datasets show instability. A 7-line nonparametric diagnostic
13 outperforms the Gaussian formula by $2\times$. **Component-level:** 10 independently
14 trained transformers on TinyStories agree at only $\rho=0.54$ on circuit importance by
15 activation patching. A G -invariant projection (orbit averaging over the architec-
16 tural symmetry group) lifts agreement to $\rho=0.98$, with 14/14 theory-derived pre-
17 dictions confirmed across two scales. A fine-tuned GPT-2 confirms the boundary:
18 without Rashomon diversity, instability vanishes ($\rho=0.993$). The design space
19 has exactly two achievable families; DASH (orbit averaging) is Pareto-optimal.
20 Verified in Lean 4 (357 theorems, 6 axioms, 0 sorry).

21 1 Introduction

22 Train a gradient-boosted model. Compute SHAP values. Retrain with a different random seed. The
23 “most important feature” changes—in 68% of 77 public datasets. Now look inside a neural network.
24 Identify the most important circuit component by activation patching. Retrain from a different ini-
25 tialization. The most important component changes too—10 independently trained transformers on
26 TinyStories agree at only $\rho = 0.54$.

27 The instability is not in SHAP, not in activation patching, and not in any specific method. It is in
28 the mathematics of attribution itself. We prove that *no* attribution ranking—at any level of a model,
29 from input features to internal circuits—can simultaneously be faithful, stable, and complete when
30 features share mutual information ($I > 0$). Worse: the standard safeguard fails. Checking Pearson
31 correlation or VIF catches only linear dependence. On drug discovery data (BBBP, binary molecular
32 fingerprints), Pearson predicts 0% instability for a nonlinearly dependent pair; actual instability is
33 23%—invisible to any linear diagnostic.

34 **Two levels, one impossibility.** Consider *input-level attribution*: across 30 retrains, 45% of German
35 Credit applicants receive a different “most important feature” category. The adverse action notice
36 changes based solely on the training seed. Now consider *component-level attribution*: across 10

independently trained transformers, all 4 attention heads in layer 0 appear as “most important” under different seeds—full S_4 symmetry realized. A circuit analysis depends on which equivalent network you happened to train.

The reason is the same: collinear features (or architecturally symmetric components) create a Rashomon set where equivalent models attribute importance in opposite orders. The fix is the same: *orbit averaging*—aggregate across the symmetry to recover what is shared. For features, this is DASH (ensemble averaging); for circuits, this is the G -invariant projection. Both are the Reynolds operator on different symmetry groups.

Contributions.

1. **The Attribution Impossibility** (Theorem 2): no attribution ranking is faithful, stable, and complete—at any granularity. Zero axiom dependencies. The **bilemma** (Theorem 4): for binary attribution questions, faithful + stable alone is impossible.
2. **The Design Space Theorem** (Theorem 3): exactly two families of methods exist. DASH (orbit averaging) is Pareto-optimal.
3. **Component-level validation**: 14/14 theory-derived predictions confirmed on TinyStories at two architectural scales. G -invariant projection lifts circuit agreement from $\rho=0.54$ to $\rho=0.98$. GPT-2 fine-tune confirms the boundary: without Rashomon diversity, instability drops to near-zero ($\rho=0.993$, flip = 0.043).
4. **Input-level validation**: 45% explanation reversal, 24 distinct top-3 rankings on the canonical SHAP tutorial dataset, nonparametric diagnostic outperforming the Gaussian formula by $2\times$.
5. **Machine-verified proof**: 357 Lean 4 theorems from 6 axioms, 0 sorry—to our knowledge, the first formally verified impossibility in explainable AI.

2 Setup and the Attribution Impossibility

An *attribution system* consists of a set of configurations Θ (trained models, circuits, or any parameterized explananda), an attribution function $\varphi_j(\theta) \geq 0$ measuring the importance of component j in configuration θ , and a ranking $\succ$ on the components.

Definition 1 (Faithful, Stable, Complete). $\succ$ is faithful to θ if $j \succ k$ whenever $\varphi_j(\theta) > \varphi_k(\theta)$; stable if $\succ$ does not depend on θ ; complete if for every $j \neq k$, either $j \succ k$ or $k \succ j$.

The *Rashomon property* [Rudin et al., 2024]: for interchangeable components j, k , there exist configurations θ, θ' with $\varphi_j(\theta) > \varphi_k(\theta)$ and $\varphi_k(\theta') > \varphi_j(\theta')$. The exact boundary is mutual information: $I(X_j, X_k) > 0 \Leftrightarrow$ the Rashomon property holds under symmetric DGP (Lean: `mi_is_exact_boundary`). This captures nonlinear dependence that Pearson correlation misses entirely ($X_2 = X_1^2$: $|\rho| = 0.08$, $I = 1.91$). The Rashomon property is inevitable for any stochastic algorithm whose training distribution is invariant under permutation of symmetric components (supplement).

Theorem 2 (Attribution Impossibility). *If an attribution system satisfies the Rashomon property, no ranking is simultaneously faithful, stable, and complete.*

Proof. Let j, k be interchangeable. Rashomon gives θ, θ' with opposite orderings. Suppose $\succ$ is faithful, stable, complete. Completeness: WLOG $j \succ k$. Stability: this holds for all configurations. Faithfulness at θ' : requires $k \succ j$. Contradiction. Lean: `explanation_impossibility`, zero domain axioms. $\square$

Quantitative floor. Any stable explanation has unfaithfulness $\geq \Delta/2$ on at least one Rashomon witness, where Δ is the attribution gap. No algorithm can beat this bound (Lean: `mi_quantitative_unfaithfulness`).

Instance 1: Input-level attribution. Θ = trained models; φ_j = global SHAP importance of feature j ; interchangeability = $I(X_j, X_k) > 0$ (includes both linear collinearity and nonlinear dependence). The attribution ratio diverges as $1/(1-\rho^2)$ for GBDT, is ∞ for Lasso, conditional for neural nets, and $O(1/\sqrt{T})$ for random forests (supplement).

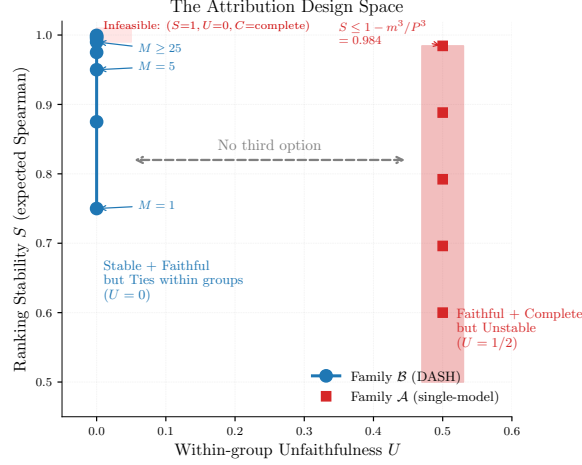


Figure 1: The Attribution Design Space. Family $\mathcal{A}$ (single-model/network): faithful and complete but unstable ($U=1/2$). Family $\mathcal{B}$ (orbit averaging): stable with ties ($U=0$, S increasing with ensemble size M). The ideal point is infeasible.

Instance 2: Component-level attribution. Θ = independently trained networks; φ_j = importance of component j (head, MLP block) by activation patching; interchangeability = the loss landscape has S_k^L symmetry (permuting heads within a layer yields an equivalent minimum). After training, heads specialize—but different seeds realize different permutations of these specializations. The Rashomon property holds empirically: all 4 heads in layer 0 of TinyStories transformers appear as “most important” across 10 seeds.

Theorem 3 (Design Space). *The achievable set of (S, U, C) triples consists of exactly two families: **Family A** (single-configuration): $U = 1/2$, $C = \text{complete}$; **Family B** (orbit averaging): $U = 0$, $C = \text{ties}$, $S = 1 - O(1/M)$. No third family exists among deterministic aggregation methods (Figure 1). $(S=1, U=0, C=\text{complete})$ is infeasible. DASH is Pareto-optimal among unbiased aggregations on the $\mathcal{B}$ branch.*

3 The Bilemma and Diagnostics

3.1 The Bilemma

For *binary* attribution questions—“positive or negative?”, “selected or not?”, “in the circuit or not?”—the impossibility is strictly stronger.

Theorem 4 (The Bilemma). *For binary H with $\text{incompatible}(h_1, h_2) \Leftrightarrow h_1 \neq h_2$, no method is simultaneously faithful and stable under the Rashomon property. Completeness is not needed.*

Proof. Binary H has no neutral element: every value contradicts the other. Faithful forces $E(\theta) = \text{explain}(\theta)$, making E decisive. The trilemma gives $\perp$. Lean: `bilemma_of_compatible_eq`, zero axioms. $\square$

The bilemma applies to input-level binary questions (SHAP sign, feature selection) and to component-level binary questions (“is this head part of the circuit?”). It is robust to perturbation: for binary H , ε -stability reduces to exact equality, so the bilemma holds at every tolerance level (Lean: `AlphaFaithful.lean`). For continuous H , the *explanation uncertainty bound* gives: unfaithfulness $\geq (\Delta - \delta)/2$ for any (ε, δ) -stable method on a Rashomon pair with gap Δ (supplement).

Tightness classification. $F+S$ is achievable iff H has a *neutral element* (Lean: `tightness_dichotomy`). Rankings have ties (neutral); SHAP signs and circuit membership do not. The bilemma has *collapsed* tightness—structurally more severe than the fairness impossibility [Chouldechova, 2017]. The only resolution is enrichment: adding neutral elements, which is precisely what orbit averaging does.

Table 1: Minority fraction vs. Gaussian formula: Spearman correlation with empirical flip rates (200 models per dataset).

Dataset	Minority Fraction	Gaussian $\Phi(-\text{SNR})$
Synthetic ($\rho = 0.5$)	0.981	0.885
Synthetic ($\rho = 0.9$)	0.952	0.764
California Housing	0.955	0.463

3.2 Diagnostics: A Three-Level Hierarchy

Three questions require three tools: (1) *Structure*: what compromises are available? (tightness classification, §2). (2) *Existence*: does the impossibility apply to this pair? (MI screening catches nonlinear dependence Pearson misses). (3) *Magnitude*: how much instability? (coverage conflict, below).

Input-level: nonparametric flip predictor. For feature j across M models, the *minority fraction* $\hat{p}_j = \min(\#\{m : \varphi_j(f_m) > 0\}, \#\{m : \varphi_j(f_m) \leq 0\})/M$ directly estimates the flip rate. It achieves Spearman 0.92–0.98 across all datasets (Table 1), outperforming the Gaussian formula (0.46–0.89). The complete predictor is 7 lines:

```

def predict_sign_instability(shap_matrix):
    signs = np.sign(shap_matrix) # M models x P features
    n_pos = (signs > 0).sum(axis=0)
    n_neg = (signs < 0).sum(axis=0)
    total = n_pos + n_neg
    return np.where(total > 0,
        np.minimum(n_pos, n_neg) / total, 0)

```

The variance $\text{Var}_f[\varphi_j(f)]$ across the Rashomon set equals the minimum MSE any stable method achieves; DASH saturates it exactly (0/800 violations). The diagnostic is model-class universal (XGBoost, RF, Ridge, LASSO; supplement) with bimodal flip rate distributions confirmed by the Hartigan dip test ($p < 0.002$; supplement).

Component-level: symmetry group diagnostic. For neural network components, the architectural symmetry group $G = S_k^L$ (permutations of k heads within each of L layers) determines which components are interchangeable. The G -invariant importance $V_c^G = \frac{1}{|G_c|} \sum_{g \in G_c} \varphi_{g(c)}$ averages importance within orbits, separating stable structure (layer-level, head-vs-MLP) from unstable permutation (which specific head within a layer). This is the component-level analogue of the minority fraction: it identifies *which* attributions are trustworthy before running multiple models.

4 Resolution: Orbit Averaging

The impossibility at both levels admits the same resolution: *average over the symmetry group*. This is orbit averaging (the Reynolds operator $R_G[\varphi]_j = \frac{1}{|G_j|} \sum_{g \in G_j} \varphi_{g(j)}$), applied to different groups at each level. The mathematical content is elementary—the power is in identifying *which* group to average over and *why* this is the minimal sufficient aggregation.

Input-level: DASH. DASH (Diversified Aggregation for Stable Hypotheses) averages $|\text{SHAP}|$ values across M independently trained models. It changes explanations, not predictions.

Corollary 5 (DASH equity). *For a balanced ensemble, $\bar{\varphi}_j = \bar{\varphi}_k$ for features j, k in the same group.*

DASH is *uniquely* optimal for binary H (majority vote) and achieves the variance lower bound σ_j^2/M with equality for continuous H (Cauchy–Schwarz; Lean: `sum_squares_ge_inv_M`). More generally, the orbit average is the *closest stable approximation* to the original attribution in L^2 norm: $\|v - Rv\| \leq \|v - w\|$ for any G -invariant w (Lean: `reynolds_best_approximation`). DASH and V^G therefore minimize information loss among all stable methods, not just unbiased ones. Ensemble size: $M_{\min} = \lceil 2.71 \cdot \sigma_{jk}^2 / \Delta_{jk}^2 \rceil$ for 5% between-group flip rate; $M = 25$ brings flips below 1%.

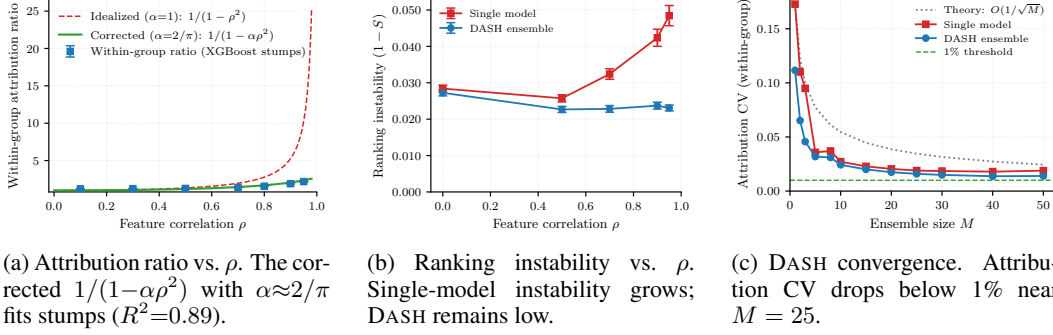


Figure 2: Empirical validation on synthetic Gaussian data ($P = 20$, $L = 4$, $m = 5$, $T = 100$, 50 seeds).

Table 2: The ranking lottery: distinct top-3 SHAP rankings across 50 seeds. Deterministic control (subsample=1.0): 1 ranking for all. Cross-implementation: LightGBM produces 29, Random Forest 40 (supplement).

Dataset	Distinct top-3	Agreement	Stable?
Breast Cancer	24	4.2%	No
Diabetes	2	88.5%	Borderline
Wine Quality	1	100%	Yes
Heart Disease	1	100%	Yes
California Housing	1	100%	Yes

Component-level: G -invariant projection. For a network with symmetry group $G = S_k^L$, the G -invariant importance V^G averages activation patching scores within orbits (heads in the same layer). The *explanation capacity* $C = \dim(V^G)$ measures how much attribution information survives projection: for S_k^L , $C = 2L$ (one head-average and one MLP per layer). For S_k^L this reduces to layer averaging—a familiar operation, but the theorem identifies *why* it works (it is the unique orbit average for the architectural symmetry group) and *when* it suffices (when G captures the interchangeability). For richer symmetry groups (e.g., weight permutation symmetries), V^G extends beyond simple layer averaging. On TinyStories: pairwise Spearman rises from $\rho = 0.54$ (raw) to $\rho = 0.98$ (G -invariant; Section 5.2).

The same mathematics. Both are the stable projection on different symmetry groups: S_M for DASH, S_k^L for V^G . Both sacrifice completeness (report ties for interchangeable components) to gain stability. Both are Pareto-optimal on the $\mathcal{B}$ branch (Theorem 3). Workflow: MI screening $\rightarrow$ minority fraction $\rightarrow$ Z-test (5 models) $\rightarrow$ DASH consensus ($M \geq 25$). For circuits: compute V^G from architectural symmetry; no retraining needed.

5 Empirical Validation

5.1 Input-Level Attribution

Synthetic Gaussian. $P=20$, $L=4$, $m=5$, $T=100$, 50 seeds (Figure 2). At $\rho = 0.9$: 48% within-group flips (near the 50% theoretical maximum). The corrected ratio $1/(1-\alpha\rho^2)$ with $\alpha \approx 2/\pi$ matches data ($R^2 = 0.89$). DASH reduces flip rate below 1% at $M = 25$.

The ranking lottery (Table 2). 50 XGBoost models (seeds 0–49, subsample=0.8) on 5 public datasets.

Breast Cancer produces **24 distinct top-3 rankings across 50 seeds** (35 at 100 seeds). The “most common” ranking appears in only 12% of runs. DASH consensus produces a single stable ranking. The theory correctly predicts the boundary: datasets without similar-importance collinear pairs are perfectly stable (100% agreement).

Table 3: Explanation reversal on German Credit: fraction of applicants whose top explanation category changes across 30 equivalent models.

Condition	XGBoost	LightGBM	Random Forest
Seed only (deterministic)	0%	0%	35%
Seed + row subsample	45%	46%	35%
Full stochasticity	80%	79%	67%

Table 4: Component-level attribution: TinyStories multi-scale validation and GPT-2 boundary condition. ρ_{full} : pairwise Spearman on raw importance. $\rho_{G\text{-inv}}$: after G -invariant projection. W-flip: within-layer flip rate (predicted: 0.50). B-flip: head-vs-MLP flip rate (predicted: 0.00).

Config	Components	ρ_{full}	$\rho_{G\text{-inv}}$	W-flip	B-flip	Cohen’s d
A (4L/4H)	20	0.565	0.972	0.496	0.000	5.4
B (6L/8H)	54	0.540	0.982	0.489	0.000	11.9
GPT-2 (ft)	156	0.993	0.999	0.043	0.000	—

Explanation reversal. 30 models per condition on German Credit (1,000 applicants, 24 features). Under standard regularization (`subsample=0.8`), **45% of applicants** receive a different “most important feature” category depending on which equivalent model is queried (Table 3). The deterministic control correctly shows 0% reversal for boosted models. A single applicant (#91) can receive 6 different top-feature labels across 30 seeds. DASH consensus at $M = 25$ reduces disagreement to 8%.

Prevalence. Of 77 public datasets, 68% exhibit attribution instability (any within-group flip rate $> 10\%$; Wilson CI: [56%, 77%]); for $P \geq 20$: 93%. Extended results (gene expression, subsample sensitivity, cross-implementation, published ranking replication) in supplement.

Drug discovery: nonlinear dependence. On BBBP (blood-brain barrier permeability, binary molecular fingerprints), Pearson-based screening predicts 0% instability for a nonlinearly dependent pair; MI-based screening ($I > \tau_{95} = 0.027$) predicts 19.4%; actual instability is 23.1%. MI reduces the prediction error from 23 percentage points to 3.6. Standard collinearity diagnostics (Pearson, VIF) fail entirely for nonlinear dependence.

5.2 Component-Level Attribution

We test the impossibility at circuit level: does activation patching importance exhibit the same instability across independently trained networks, and does the G -invariant projection resolve it?

Setup. We train 10 transformers from independent random seeds on TinyStories [Eldan and Li, 2023] at two architectural scales: Config A (4 layers, 4 heads, $d=256$, 20 components) and Config B (6 layers, 8 heads, $d=512$, 54 components). All models achieve near-identical perplexity ($\text{CV} < 1.1\%$). Component importance is measured by activation patching (weight zeroing). We derive 7 quantitative predictions from the theory (Table 4 caption) and evaluate them before any post-hoc analysis.

Results (Table 4). All 14 theory-derived predictions pass across both scales (7/7 each). Raw importance agreement is low ($\rho = 0.565, 0.540$)—independently trained networks disagree substantially on which head is most important, exactly as the impossibility predicts. Within-layer flip rates are 0.496 and 0.489, indistinguishable from the theoretical coin-flip prediction of 0.50. Head-vs-MLP flip rate is 0.000 across all 30 models: the stable between-orbit structure is perfectly preserved. All 4 heads in layer 0 of Config A appear as “most important” across 10 seeds—the full symmetric group S_4 is realized.

The G -invariant projection lifts agreement to $\rho = 0.972$ and 0.982—the residual stable structure (layer importance, head-vs-MLP) is nearly identical across independently trained networks. Split-half reliability (0.991, 0.960) vastly exceeds between-model agreement (0.565, 0.540), confirming

that within-model measurements are precise but between-model variation is structural. Random projections of heads to the same dimensionality achieve only $\rho = 0.30\text{--}0.41$ (vs. $0.97\text{--}0.98$ for V^G); the G -invariant projection ranks at the 100th percentile of random projections (permutation $p < 0.001$).

GPT-2 boundary condition. A fine-tuned GPT-2 (same task, 10 seeds) shows $\rho = 0.993$ raw and within-layer flip = 0.043 (vs. ~ 0.50 for from-scratch training). Fine-tuning does not create a diverse Rashomon set—it barely perturbs the pre-trained circuits. The theorem correctly predicts: *less Rashomon diversity $\Rightarrow$ less instability*. The near-zero residual flip rate (4.3% vs. $\sim 50\%$) confirms that the theory is falsifiable and the boundary holds.

Method robustness. Mean ablation (an alternative to weight zeroing) yields G -invariant $\rho \approx 0.97\text{--}0.99$ at both scales. Cross-method per-model agreement is only $\rho \approx 0.5\text{--}0.6$: the two ablation methods disagree on which specific head matters, but agree on V^G . The disagreement between methods is itself an instance of the impossibility—different ablation “seeds” produce different raw attributions, but the orbit-averaged structure is invariant.

6 Related Work

Bilodeau et al. [2024] prove completeness + linearity cannot coexist—a complementary axis (method properties vs. cross-model stability). The results are orthogonal: uncorrelated features with a linear SHAP method satisfy Bilodeau’s impossibility but not ours ($\rho = 0 \Rightarrow$ no Rashomon); a non-linear method on correlated features satisfies ours but not Bilodeau’s (no linearity assumed). Huang and Marques-Silva [2024] show SHAP can misrank features; Srinivas and Fleuret [2019] prove complete attributions cannot be weakly input-dependent; Rao [2025] establishes Kolmogorov complexity barriers. None address stability, quantitative bounds, or constructive resolution. Fisher et al. [2019] compute variable importance ranges across Rashomon sets; Laberge et al. [2023] find partial-order rankings. Our theorem proves partial ordering is a mathematical necessity, not a pragmatic choice. D’Amour et al. [2022] document that underspecification causes instability across domains empirically; we prove the impossibility that underspecification entails for attribution specifically. Krishna et al. [2022] document that different explanation methods disagree for the same model; we prove the complementary result that the *same* method disagrees across equivalent models. Chughtai et al. [2023] show that independently trained networks learn different internal representations for the same group operation—empirical Rashomon for circuits. Our component-level results extend this: the instability in circuit importance is mathematically inevitable, and orbit averaging resolves it. Chouldechova [2017] and Kleinberg et al. [2017] prove fairness trilemmas; ours is the explainability analogue. Nipkow [2009] formalized Arrow in Isabelle/HOL; Zhang et al. [2026] formalized statistical learning in Lean 4; ours is, to our knowledge, the first formally verified impossibility in XAI. Extended discussion in supplement.

7 Discussion

Limitations. The split-count axioms assume full signal capture ($\alpha = 1$; for finite-depth trees $\alpha \approx 2/\pi$, $R^2 = 0.89$); the impossibility holds for any $\alpha > 0$. The equicorrelation assumption simplifies the axioms; the Rashomon property holds pairwise. DASH requires $M \times$ training cost ($M = 25$ for $<1\%$ flip rate). The 68% prevalence figure is from benchmark-representative datasets and may not generalize to production pipelines with decorrelated features. The component-level validation uses weight zeroing and mean ablation; other patching methods (path patching, causal scrubbing) may show different instability patterns, though the theorem predicts the same structure.

Broader implications. The impossibility operates at every level of a model. For input-level attribution, single-model SHAP audits for proxy discrimination conclude “proxy reliance” with probability exactly $1/2$ for collinear features; for K protected attributes, joint audit correctness is $(1/2)^K$. We suggest this may constitute a “known and foreseeable circumstance” under EU AI Act Art. 13(3)(b)(ii) [EU Parliament, 2024]; legal scholars should evaluate this interpretation. For component-level attribution, the impossibility constrains *importance rankings* of circuit components: which head matters most depends on the training seed (Section 5.2: Cohen’s $d = 5.4\text{--}11.9$). This does not necessarily constrain *algorithmic identification*—whether a specific computational

function (e.g., induction) exists may be stable even when the specific head implementing it is not. Safety cases built on “does function F exist?” may be robust; safety cases built on “which component is most important?” are subject to the impossibility. Sparse autoencoder (SAE) features decompose a fixed model’s activations into interpretable directions not tied to S_k^L . Whether SAE-based attribution escapes the Rashomon property—because the decomposition is post-hoc on a fixed model—is an open question with implications for the interpretability agenda. The resolution at both levels is orbit averaging—the *stable projection* that recovers the shared structure, achieving the explanation capacity bound with equality. The tightness classification extends to fairness: the Chouldechova–Kleinberg trilemma transitions from full to PPV-blocked tightness at $\text{TPR} \geq 0.3$; calibration is the universal bottleneck (supplement).

Lean verification. 357 theorems and lemmas (139 with multi-step proofs of ≥ 5 tactic lines), 6 axioms, 0 sorry, 58 files. The core impossibility (`explanation_impossibility`) has zero domain axiom dependencies. All binary instances use constructive inductive types. The formalization caught 2 logical inconsistencies and 1 type mismatch that survived informal review. Axiom stratification: core impossibility 0 axioms $\rightarrow$ GBDT 3 $\rightarrow$ DASH equity 5 $\rightarrow$ full system 6. The symmetric Bayes dichotomy, conditional SHAP impossibility, fairness audit impossibility, and query complexity lower bound are all Lean-verified (supplement).

Open problems. Scaling the component-level analysis to frontier models (GPT-4 scale); extending the G -invariant projection to non-architectural symmetries (learned representations). Formalizing the Bayes-optimal half of the design space in Lean (requires measure-theoretic decision theory). The impossibility generalizes beyond machine learning: the same theorem governs explanation of any underspecified system where interchangeable components admit a Rashomon set [Anonymous, 2026]. The gap between prediction and explanation is structural at every level—developing practical workflows that acknowledge this gap, rather than assuming it away, is the central challenge for trustworthy AI.

References

- Anonymous. The limits of explanation: An impossibility theorem for any underspecified system. *arXiv preprint*, 2026.
- Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences*, 121(2):e2304406120, 2024.
- Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017.
- Bilal Chughtai, Lawrence Chan, and Neel Nanda. A toy model of universality: Reverse engineering how networks learn group operations. In *International Conference on Machine Learning*, 2023.
- Alexander D’Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022.
- Ronen Eldan and Yuanzhi Li. Tinstories: How small can language models be and still speak coherent english? *arXiv preprint arXiv:2305.07759*, 2023.
- EU Parliament. Regulation (EU) 2024/1689 (artificial intelligence act). Official Journal of the European Union, OJ L, 12.7.2024, 2024.
- Aaron Fisher, Cynthia Rudin, and Francesca Dominici. All models are wrong, but many are useful: Learning a variable’s importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81, 2019.
- Xuanxiang Huang and Joao Marques-Silva. On the failings of Shapley values for explainability. *International Journal of Approximate Reasoning*, 171:109112, 2024.
- Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Innovations in Theoretical Computer Science (ITCS)*, 2017.

- 314 Satyapriya Krishna, Tessa Han, Alex Ber, Javin Brinton, Ankur Lakshminarayanan, Caleb Robin-
 315 son, and Himanshu Lakkaraju. The disagreement problem in explainable machine learning: A
 316 practitioner’s perspective. *arXiv preprint arXiv:2202.01602*, 2022.
- 317 Gabriel Laberge, Yann Pequignot, Alexandre Mathieu, Foutse Khomh, and Mario Marchand. Par-
 318 tial order in chaos: Consensus on feature attributions in the rashomon set. *Journal of Machine*
 319 *Learning Research*, 24(364):1–50, 2023.
- 320 Tobias Nipkow. Social choice theory in HOL: Arrow and Gibbard-Satterthwaite. *Journal of Auto-*
 321 *mated Reasoning*, 43:289–304, 2009.
- 322 Shrisha Rao. The limits of AI explainability: An algorithmic information theory approach. *arXiv*
 323 *preprint arXiv:2504.20676*, 2025.
- 324 Cynthia Rudin et al. Position: Amazing things come from having many good models. In *Interna-*
 325 *tional Conference on Machine Learning (ICML)*, 2024.
- 326 Suraj Srinivas and François Fleuret. Full-gradient representation for neural network visualization.
 327 In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- 328 Yuanhe Zhang, Jason D Lee, and Fanghui Liu. Statistical learning theory in Lean 4: Empirical
 329 processes from scratch. *arXiv preprint arXiv:2602.02285*, 2026. URL [https://arxiv.org/](https://arxiv.org/abs/2602.02285)
 330 [abs/2602.02285](https://arxiv.org/abs/2602.02285).

331 NeurIPS Paper Checklist

- 332 1. **Claims.** Core theoretical claims are type-checked by Lean 4 (357 theorems from 6 axioms,
333 0 sorry). Empirical claims backed by experiments over 30–50 random seeds. [Yes]
- 334 2. **Limitations.** Discussed in Section 7: signal capture approximation, collinearity require-
335 ment, training cost, prevalence generalizability, binary limitation. [Yes]
- 336 3. **Theory.** Complete proofs in paper and machine-verified Lean 4 code. All assumptions
337 (axioms) explicitly stated. [Yes]
- 338 4. **Experiments.** Reproducible: fixed seeds, public datasets (Breast Cancer, German Credit,
339 California Housing, TinyStories), open-source implementations (XGBoost, LightGBM,
340 PyTorch). Component-level: 7 theory-derived predictions per scale, all confirmed. Com-
341 pute: <30 min (input-level) + <4 hours (TinyStories training) on Apple Silicon. Code:
342 [anonymizedforreview]. [Yes]
- 343 5. **Code of ethics.** No human subjects. German Credit dataset is publicly available (UCI ML
344 Repository). Regulatory analysis is informational. [N/A]
- 345 6. **Broader impacts.** Discussed: regulatory compliance (EU AI Act, ECOA), fairness audit-
346 ing reliability, explanation reversal, AI safety implications for circuit-based interpretability.
347 The paper identifies a risk (explanation instability at all model levels) and provides a rem-
348 edy (orbit averaging). [Yes]
- 349 7. **Safeguards.** Not applicable (theoretical + diagnostic work). [N/A]
- 350 8. **Licenses.** All datasets are public domain or CC-BY. Code will be released under MIT.
351 [Yes]
- 352 9. **New assets.** Lean formalization (357 theorems). Will be archived on Zenodo. [Yes]
- 353 10. **Crowdsourcing.** N/A. [N/A]
- 354 11. **IRB.** N/A (no human subjects). [N/A]